

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

PACIFIC BIOSCIENCES OF CALIFORNIA, INC.,
Petitioner,

v.

PERSONAL GENOMICS TAIWAN, INC.,
Patent Owner.

IPR2020-01163
Patent 7,767,441 B2

Before ERICA A. FRANKLIN, SUSAN L. C. MITCHELL, and
MICHAEL VALEK, *Administrative Patent Judges*.

MITCHELL, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

A. *Background and Summary*

On June 22, 2020, Pacific Biosciences of California, Inc. (“Petitioner”) filed a Petition (Paper 1, “Pet.”) requesting *inter partes* review of claims 1–6, 9, and 43–58 (“the challenged claims”) of U.S. Patent No. 7,767,441 B2 (Ex. 1001, “the ’441 patent”). On October 22, 2020, Personal Genomics Taiwan, Inc. (“Patent Owner”) filed a Preliminary Response to the Petition. Paper 13 (“Prelim. Resp.”). On January 19, 2021, we granted institution of an *inter partes* review. See Paper 15 (“Dec.”), 3–4, 38.

Patent Owner filed a Response on April 13, 2021, Paper 19 (“Resp.”), and Petitioner filed a Reply on July 6, 2021. Paper 21 (“Reply”). Patent Owner filed its Sur-Reply on August 17, 2021. Paper 22 (“Sur-Reply”). An oral hearing was held on October 21, 2021. Paper 30 (“Tr.”).

This is a Final Written Decision under 35 U.S.C. § 318(a) as to the patentability of the challenged claims on which we instituted trial. Based on the complete record before us, we determine that Petitioner has shown, by a preponderance of the evidence, that claims 1–6, 9, and 43–58 are unpatentable.

B. *Real Parties in Interest*

Each party identifies itself as the real party-in-interest. Pet. 4; Paper 3, 2.

C. *Related Matters*

The parties identify an ongoing court proceeding involving the ’441 patent: *Personal Genomics Taiwan, Inc. v. Pacific Biosciences of*

California, Inc., No. 19-cv-1810-LPS (D. Del. filed Sept. 26, 2019). Pet. 4, Paper 3, 2.

On June 27, 2020, Petitioner filed a second petition for *inter partes* review that challenges claims 1, 2, 6, 7, 10–22, 24, and 27–36 of the ’441 patent. *See Pacific Biosciences of California, Inc. v. Personal Genomics Taiwan, Inc.*, IPR2020-01200, Paper 1 (PTAB June 27, 2020) (“1200 Petition”). We granted institution of an *inter partes* review as to these challenged claims. *See* IPR2020-01200, Paper 15, 3–4, 35 (PTAB January 19, 2021).

D. The ’441 Patent

The ’441 patent, titled “Bioassay System Including Optical Detection Apparatuses, and Method for Detecting Biomolecules,” generally relates to a bioassay system including a plurality of optical detection apparatuses that can sense the existence of a fluorophore on a single molecule. *See* Ex. 1001, code (54), 3:55–61. The ’441 patent describes the need for a device that should be “capable of sequencing single molecules to avoid the known difficulty of asynchrony in both the amplification (e.g., drift between the sequences of ideally clonal templates) and sequencing (e.g., dephasing of the stepwise sequencing reactions amongst the sequencing templates) steps of clustered sequencing methods.” *Id.* at 1:67–2:6. The ’441 patent states that “[t]he bioassay system provided by the invention is capable of large-scale parallel sequencing reactions, i.e., simultaneously sequencing a large number of different nucleic acid templates. Each sequencing reaction uses a single molecule as the template (i.e., single molecule sequencing).” *Id.* at 2:13–18. The invention provides “direct correspondence of linker sites to which the nucleic acids being detected are attached . . . and one or more

detecting units (e.g., light detectors), and in part, on the short distance between the linker sites and the detecting units.” *Id.* at 2:21–27.

Figure 2 of the ’441 patent is reproduced below.

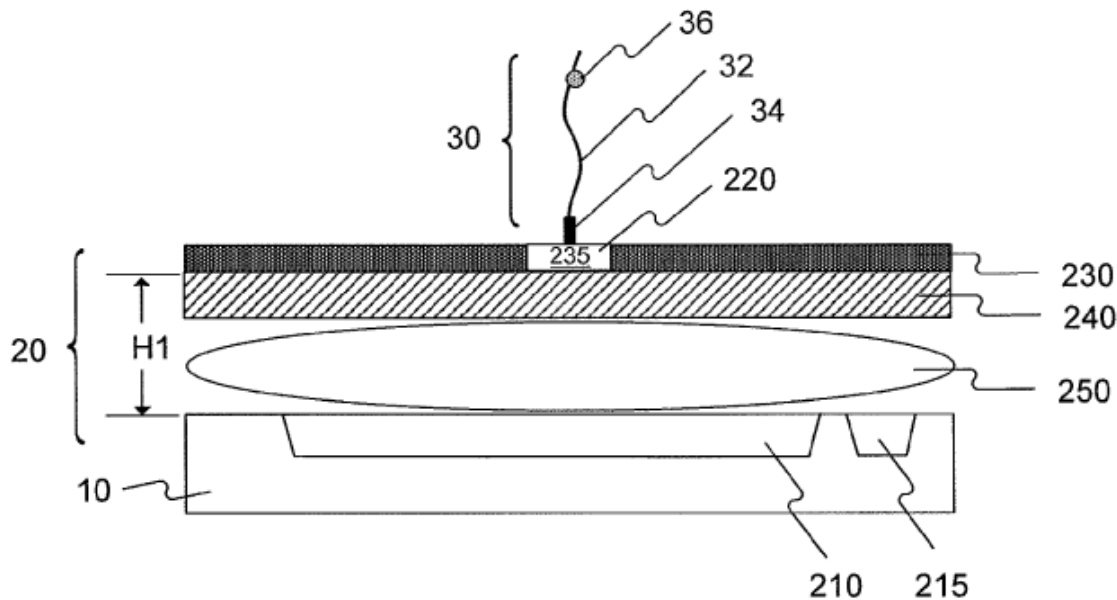


Fig. 2

Figure 2 above illustrates an optical detection apparatus. *Id.* at 3:24–26. Optical detection apparatus 20 includes light detector 210 formed on substrate 10. *Id.* at 4:52–60. Light detector 210 may alternatively include one or more photoconductive photon detectors, photovoltaic photon detectors, or photodiodes. *Id.* at 5:1–11. Optical detection apparatus 20 may also include blind sheet 230 over light detector 210, including pinhole 235 having a diameter of less than 1,000 nanometers. *Id.* at 5:12–22. The ’441 patent further states that “linker site 220 formed proximate to pinhole 235 may be spaced apart from light detector 210 by a distance H1 of less than or equal to 100 micrometers.” *Id.* at 5:25–28. Optical detection apparatus 20 may optionally include filter layer 240 and microlens 250 between light

detector 210 and blind sheet 230. *Id.* at 5:31–33. Filter layer 240 may include one or more transparent sublayers. *Id.* at 5:36–38.

The '441 patent states that “linker site 220 may be treated to affix a single biomolecule 30 thereto.” Ex. 1001, 5:49–50. For example, biomolecule 30 may include single stranded DNA molecule 32, affixed to linker site 220 with end link primer 34. *Id.* at 5:50–54. DNA molecule 32 may be labeled with fluorophore 36. *Id.* at 5:54–55. The '441 patent also states that “[w]hen excited by excitation light of a first wavelength λ_1 , fluorophore 36 may emit fluorescent light of a second wavelength λ_2 .” *Id.* at 5:55–57. Furthermore, “[l]ight detector 210 then detects the fluorescent light emitted from fluorophore 36, so as to identify the type of base that fluorophore 36 is attached to, thereby sequentially determining the sequence of DNA molecule 32.” *Id.* at 5:60–64.

E. Challenged Claims

Petitioner challenges claims 1–6, 9, and 43–58 of the '441 patent. Claims 1, 43, 48, and 53–55 are the only challenged independent claims. Challenged claims 2–6 and 9 depend directly or indirectly from claim 1, challenged claims 44–47 depend directly or indirectly from claim 43, challenged claims 49–52 depend directly or indirectly from claim 48, and challenged claims 56–58 depend directly or indirectly from claim 55.

Claim 1 is illustrative and recites:

1. An apparatus for identifying a single biomolecule, comprising:
 - a substrate having a light detector; and
 - a linker site formed over the light detector, the linker site being treated to affix the biomolecule to the linker site;

wherein the linker site is proximate to the light detector and is spaced apart from the light detector by a distance of less than or equal to 100 micrometers.

Ex. 1001, 26:11–18.

F. Prior Art and Asserted Grounds of Unpatentability

Petitioner asserts that claims 1–6, 9, and 43–58 would have been unpatentable based on the following grounds:

Claim(s) Challenged	35 U.S.C. §	Reference(s)/Basis
1–5, 9, 48, 52–56	102 ¹	Choumane ²
1–5, 9, 48, 52–56	103(a)	Choumane, Weisbuch ³
6, 9, 49–51, 53, 54	103(a)	Choumane, Weisbuch
43–48	103(a)	Choumane in view of the skill of a POSA (Person of Ordinary Skill in the Art)
57, 58	103(a)	Choumane in view of the knowledge of a POSA

Petitioner submits the Declarations of Dean P. Neikirk, Ph.D. in support of its positions. *See* Ex. 1003; Ex. 1059. Patent Owner submits the Declaration of Timothy D. Harris, Ph.D. *See* Ex. 2008.

¹ The Leahy-Smith America Invents Act (“AIA”) included amendments to 35 U.S.C. §§ 102 and 103. Because the ’441 patent issued from an application filed before the effective date of these amendments, we apply the pre-AIA versions of §§ 102 and 103.

² Choumane et al., WO 2007/045755 A1, published Apr. 26, 2007 (Ex. 1005, “Choumane”). We rely on the certified English translation filed in the record as Ex. 1006.

³ Weisbuch et al., US 2002/0182716 A1, published Dec. 5, 2002 (Ex. 1007, “Weisbuch”).

II. ANALYSIS

A. Principles of Law

1. Burden

In an *inter partes* review, the burden of proof is on the Petitioner to show that the challenged claims are unpatentable, and that burden never shifts to the patentee. *See* 35 U.S.C. § 316(e); *In re Magnum Oil Tools Int’l, Ltd.*, 829 F.3d 1364, 1375 (Fed. Cir. 2016) (citing *Dynamic Drinkware, LLC v. Nat’l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015)).

2. Anticipation

To establish anticipation, each and every element in a claim, arranged as recited in the claim, must be found in a single prior art reference. *Net MoneyIN, Inc. v. VeriSign, Inc.*, 545 F.3d 1359, 1369 (Fed. Cir. 2008); *Karsten Mfg. Corp. v. Cleveland Golf Co.*, 242 F.3d 1376, 1383 (Fed. Cir. 2001). “A reference anticipates a claim if it discloses the claimed invention ‘such that a skilled artisan could take its teachings in combination with his own knowledge of the particular art and be in possession of the invention.’” *In re Graves*, 69 F.3d 1147, 1152 (Fed. Cir. 1995) (citation and emphasis omitted). Moreover, “it is proper to take into account not only specific teachings of the reference but also the inferences which one skilled in the art would reasonably be expected to draw therefrom.” *In re Preda*, 401 F.2d 825, 826 (CCPA 1968); *see Eli Lilly and Co. v. Los Angeles Biomedical Res. Inst. at Harbor-UCLA Medical Center*, 849 F.3d 1073, 1074–75 (Fed. Cir. 2017).

An apparatus claim covers “what a device *is*, not what a device *does*.” *ParkerVision, Inc v. Qualcomm Inc.*, 903 F.3d 1354, 1361 (Fed. Cir. 2018) (citation omitted).

As a result, “[a]n invention need not *operate* differently than the prior art to be patentable, but need only *be* different”—or, rather, “*unobviously* different.” A corollary of these principles is that an apparatus that is “capable of” performing certain functions may be anticipated by or obvious in view of a prior art apparatus that can likewise perform these functions. Indeed, “depending on the claims, ‘an accused device may be found to infringe if it is reasonably capable of satisfying the claim limitations, even though it may also be capable of noninfringing modes of operation. Similarly, a prior art reference may also anticipate or render obvious an apparatus claim—depending on the language of the claim—if the reference discloses an apparatus that is reasonably capable of operating so as to meet the claim limitations, even if it does not meet the claim limitations in all modes of operation.

Id. (citations omitted).

3. *Obviousness*

A patent claim is unpatentable under 35 U.S.C. § 103 if the differences between the claimed subject matter and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains (“POSA” or “POSITA”). *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) objective evidence of nonobviousness.⁴ *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966). In determining obviousness when all elements of a claim are found in various pieces of prior art, “the factfinder must further

⁴ Patent Owner also does not present any objective evidence of nonobviousness.

consider the factual questions of whether a person of ordinary skill in the art would be motivated to combine those references, and whether in making that combination, a person of ordinary skill would have had a reasonable expectation of success.” *Dome Patent L.P. v. Lee*, 799 F.3d 1372, 1380 (Fed. Cir. 2015); *see also WMS Gaming, Inc. v. Int’l Game Tech.*, 184 F.3d 1339, 1355 (Fed. Cir. 1999) (“When an obviousness determination relies on the combination of two or more references, there must be some suggestion or motivation to combine the references.”). “Both the suggestion and the expectation of success must be founded in the prior art, not in the applicant’s disclosure.” *In re Dow Chemical Co.*, 837 F.2d 469, 473 (Fed. Cir. 1988).

An obviousness analysis “need not seek out precise teachings directed to the specific subject matter of the challenged claim, for a court can take account of the inferences and creative steps that a person of ordinary skill in the art would employ.” *KSR*, 550 U.S. at 418; *see In re Translogic Tech, Inc.*, 504 F.3d 1249, 1259 (Fed. Cir. 2007). In *KSR*, the Supreme Court also stated that an invention may be found obvious if trying a course of conduct would have been obvious to a POSITA:

When there is a design need or market pressure to solve a problem and there are a finite number of identified, predictable solutions, a person of ordinary skill has good reason to pursue the known options within his or her technical grasp. If this leads to the anticipated success, it is likely the product not of innovation but of ordinary skill and common sense. In that instance the fact that a combination was obvious to try might show that it was obvious under § 103.

550 U.S. at 421. “*KSR* affirmed the logical inverse of this statement by stating that § 103 bars patentability unless ‘the improvement is more than the predictable use of prior art elements according to their established

functions.”” *In re Kubin*, 561 F.3d 1351, 1359–60 (Fed. Cir. 2009) (citing *KSR*, 550 U.S. at 417).

We analyze the asserted grounds of unpatentability in accordance with the above-stated principles.

B. Level of Ordinary Skill in the Art

We consider the asserted grounds of unpatentability in view of the understanding of a person of ordinary skill in the art. *KSR*, 550 U.S. at 399 (stating that obviousness is determined against the backdrop of the scope and content of the prior art, the differences between the prior art and the claims at issue, and the level of ordinary skill in the art). Factual indicators of the level of ordinary skill in the art include “the various prior art approaches employed, the types of problems encountered in the art, the rapidity with which innovations are made, the sophistication of the technology involved, and the educational background of those actively working in the field.” *Jacobson Bros., Inc. v. U.S.*, 512 F.2d 1065, 1071 (Ct. Cl. 1975); *see also Orthopedic Equip. Co. v. U.S.*, 702 F.2d 1005, 1011 (Fed. Cir. 1983) (quoting with approval *Jacobson Bros.*).

In our Decision on Institution, we applied Petitioner’s proposed definition of one skilled in the art as an individual with “(1) at least [a] Bachelor of Arts or Bachelor of Science degree in biology, biochemistry, applied physics, or a similar degree,” and (2) “at least 1-2 years of experience in the field of optics and sensors.” Dec. 8.

In the Response, Patent Owner argues that the person of ordinary skill in the art should also have 1–2 years of experience in “molecular sequencing, including handling and sequencing of nucleic acids.” Resp. 2 (citing Ex. 2008 ¶ 23). Patent Owner supports this contention by citing to

various discussions throughout the Specification of the '441 patent referring to the sequencing of nucleic acids. *See id.* at 2–3 (citing Ex. 2008 ¶¶ 16–22). Patent Owner also contends that the language recited in the claims supports this definition. *Id.*

Patent Owner argues Petitioner's expert, Dr. Neikirk, does not have the corresponding experience in molecular sequencing. *See* Resp. 3. Accordingly, Patent Owner contends that Dr. Neikirk's opinions relating to molecular sequencing in the claims and the prior art, "should be accorded little or no weight." *Id.* at 3–4.

Petitioner responds that "the alleged invention of the '441 patent is simply a general purpose molecular sensor." Reply 2. Petitioner asserts that the Specification describes several other applications for the sensor besides sequencing, and that only some of the claims require sequencing. *See id.* (citing Ex. 1001, 1:11–22, 7:58–62, 15:39–17:33). Accordingly, Petitioner asserts that no specialized sequencing knowledge is required. *Id.*

We find Petitioner has the better argument. The claims at issue in this *inter partes* review involve a molecular sensor, its components, or a method of using or manufacturing the sensor. *See* Ex. 1001, 26:11–37, 27:64–30:6. An in-depth understanding of molecular sequencing would not be required to understand these claims, the '441 patent, the prior art asserted against these claims, or the problems solved by the claimed apparatus and methods. Therefore, we will continue to apply Petitioner's definition of one of skill in the art as set forth above.

Moreover, even under Patent Owner's proposal for the level of ordinary skill, we find that Dr. Neikirk is qualified to offer the opinions in his declaration. "There is . . . no requirement of a perfect match between the

expert’s experience and the relevant field” and one need “not be a person of ordinary skill in the art in order to testify as an expert under Rule 702” so long as they are “qualified in the pertinent art.” PTAB Consolidated Trial Practice Guide (Nov. 2019) 34 (citing precedent). Here, the record demonstrates that Dr. Neikirk is qualified in the art of molecular sensors. *See* Ex. 1003 ¶¶ 4–20; Ex. 1004 (describing Dr. Neikirk’s technical background and expert qualifications). Therefore even if one of ordinary skill would have had experience in molecular sequencing, as Patent Owner proposes, it does not follow that all of the testimony in Dr. Neikirk’s declarations should be accorded “little or no weight.”

C. Claim Construction

We construe claims using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. § 282(b). 37 C.F.R. § 42.100 (2019). Therefore, we construe the challenged claims under the framework set forth in *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–19 (Fed. Cir. 2005) (en banc). Under this framework, claim terms are given their ordinary and customary meaning, as would be understood by a person of ordinary skill in the art, at the time of the invention, in light of the language of the claims, the specification, and the prosecution history of record. *Id.* Only those terms that are in controversy need be construed and only to the extent necessary to resolve the controversy. *See Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co. Matal*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (citing *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999)).

At the preliminary stage of the proceeding, Petitioner did not offer any express claim constructions for any claim term, *see* Pet. 9, but Patent Owner

did, *see* Prelim. Resp. 19–30. Patent Owner asserted that the preambles of the claims calling for a “identifying a single biomolecule” are limiting, and each of the claims requires a “linker site” to which a “single biomolecule” is affixed. *Id.*

In our Institution Decision, we did not determine whether the preamble limits the claims to identifying a single biomolecule because we found that, on the preliminary record, Choumane discloses an apparatus for identifying a single biomolecule. Dec. 9–10. We did interpret, however, the claims phrase “to affix *the* biomolecule” as referring to affixing “a single biomolecule.” *Id.* at 10 n.8.

We also addressed the construction of the claim limitation “a linker site formed over the light detector, the linker site being treated to affix the biomolecule to the linker site.” We stated that “[a]lthough the challenged apparatus claims are ‘capable of’ affixing a single biomolecule to the linker site, the structure of the apparatus is not limited to being capable of affixing *only* a single biomolecule to the linker site.” *Id.* at 10–11 (citing claim 30).

In the Response, Patent Owner again argues that the preambles of challenged claims 1, 48, 53, and 54 limit the claims to “an apparatus for identifying a single biomolecule.” *See* Resp. 4–16. Patent Owner also argues that the challenged claims require “a linker site” treated to affix a *single* biomolecule to be identified by the apparatus. *Id.* at 11–12. Patent Owner further argues that the claim phrase “linker site formed over the light detector” at least requires that “the linker site be configured directly or indirectly on top of, upon, or above” the light detector. *Id.* at 16–18.

We discuss below each of these claim terms or phrases and the parties’ arguments with respect to the same.

1. Preamble Effect – Identifying a Single Biomolecule

Patent Owner argues that the preambles of independent claims 1, 48, 53, and 54, limit the claims to identifying a single biomolecule. *See* Resp. 4–16. In support of the assertion, Patent Owner first argues that the preamble provides antecedent basis for claim term “the biomolecule” referenced in the bodies of each of these independent claims. *Id.* at 5–6 (citing *CW Zumbiel Co., Inc. v. Kappos*, 702 F.3d 1371, 1385 (Fed. Cir. 2012)). By relying on the preamble for antecedent basis, Patent Owner argues that the claims use both the preambles and bodies of the claims to define the invention itself. *See id.* at 6 (citing *Bell Comms. v. Vitalink Comms. Corp.*, 55 F.3d 615, 620 (Fed. Cir. 1995)).

Second, Patent Owner argues that the preambles recite a fundamental characteristic of the invention, i.e., identifying a single biomolecule, that is necessary to define the invention. Resp. 6–9 (citing *On Demand Machine Corp. v. Ingram Indus., Inc.*, 442 F.3d 1331, 1343 (Fed. Cir. 2006) (finding the preamble at issue there limited the claims because it stated the framework of the invention and was fundamental to the invention)).

Patent Owner argues that, similar to *On Demand*, “the specification of the ’441 patent describes identifying a single biomolecule as fundamental to the invention.” Resp. 8. Specifically, Patent Owner argues that “[a]ll of the apparatuses in the figures of the ’441 patent are structures for identifying a single biomolecule,” and all of the Examples “describe apparatuses for detecting a single biomolecule and methods for making and using them, including for sequencing a nucleic acid.” *Id.* at 8–9 (citing Ex. 2008 ¶¶ 37–38). Patent Owner argues that a person of ordinary skill in the art would have understood these disclosures, combined with the Summary and

Detailed Description of the Specification, “to communicate that an apparatus for identifying a single biomolecule is fundamental to the invention.” *Id.* (citing Ex. 2008 ¶ 39).

Patent Owner also asserts that a “single” biomolecule as used in the ’441 patent and required by the claims “is an exclusive word referring to a count of only one.” Resp. 9. Patent Owner concludes that “in view of the specification and the claims, the plain and ordinary meaning of ‘[a]n apparatus for identifying a single biomolecule’ to a POSA is a structure for identifying an individual biomolecule, as opposed to, for example, multiple copies of a biomolecule in an ensemble.” *Id.* at 10 (citing Ex. 2008 ¶¶ 40–43; Ex. 1001, 2:16–18, 2:30–31, 4:2–4, 5:49–50, 12:36–58, 15:11–13).

Petitioner responds that the preamble recites “identifying a single molecule,” and not “detecting a signal from only a single molecule.” Reply 3. Petitioner asserts that identifying a single molecule can be performed by copying a single target molecule and analyzing the signal from a collection of identical molecules. *See id.* Petitioner asserts that Patent Owner’s expert confirms this interpretation of “identify” with respect to “the well-known Illumina sequencing platform,” which analyzes signal “from a collection of identical molecules that have been amplified on a surface.” *Id.* at 4.

Petitioner asserts that the dependent claims also support its interpretation. *See* Reply 5–8. Specifically, dependent claim 26, recites “wherein the nucleic acid is amplified at the linker site before nucleic acid sequencing.” *Id.* at 5. Petitioner asserts that amplifying the nucleic acid “expressly contemplates detecting a signal from *multiple* molecules of the same species.” *Id.* (citing Ex. 1003 ¶ 60). Petitioner also refers to claim 30,

which recites “affixing ***one or more*** biomolecule to the linker site of the apparatus of claim 1.” *Id.* at 7.

Petitioner further contends that the Specification, prosecution history, and extrinsic evidence support its claim interpretation. Reply at 8–12. First, Petitioner argues that the ’441 patent cites to multiple publications for describing “Sequencing Modalities.” *Id.* at 8–11 (citing Ex. 1001, 12:37–40, 13:2–4). Petitioner contends that Dr. Harris acknowledges these references disclose “techniques in which one detects signal from multiple molecules of the same species.” *Id.* at 9–11 (citing Ex. 1058, 86:16–25, 79:14–80:10, 83:2–84:15, 88:12–17, 77:9–24).

Second, Petitioner contends that the ’441 patent claims the benefit of provisional application No. 61/036,652, and the provisional application “expressly claims a process ‘wherein a amplification process can be included after said nucleic acid binded to said binding site.’” Reply at 11 (citing Ex. 1056, 40). Finally, Petitioner contends that the extrinsic evidence supports its position based on Dr. Harris’ previous work at Helicos. *Id.* at 11–12 (citing Ex. 2008 ¶¶ 8–11). Specifically, Petitioner asserts that Helicos previously described an alleged infringing “Illumina platform as carrying out ‘single molecule sequencing,’” despite the Illumina platform sequencing multiple copies of a molecule together. *See id.* at 12 (citing Ex. 1054 ¶¶ 39, 137; Ex. 1058, 107:6–9).

In construing claim terms, we start with the language of the claims themselves. *See Phillips*, 415 F.3d at 1314 (stating “the claims themselves provide substantial guidance as to the meaning of particular claim terms”). A preamble to a claim should be construed as limiting the scope of the claim “[i]f the claim preamble, when read in the context of the entire claim, recites

limitations of the claim, or, if the claim preamble is ‘necessary to give life, meaning, and vitality’ to the claim.” *Pitney Bowes, Inc. v. Hewlett-Packard Co.* 182 F.3d 1298, 1305 (Fed. Cir. 1999). The Federal Circuit has also stated:

[D]ependence on a particular disputed preamble phrase for antecedent basis may limit claim scope because it indicates a reliance on both the preamble and claim body to define the claimed invention. Likewise, then the preamble is essential to understand limitations or terms in the claim body, the preamble limits claim scope.

Catalina Marketing Internat’l v. Coolsavings.com, Inc., 289 F.3d 801, 808 (Fed. Cir. 2002) (citing *Bell*, 55 F.3d at 620; *Pitney Bowes*, 182 F.3d at 1306).

Here, we find that the preamble reciting “a single biomolecule” for each independent claim provides antecedent basis for the claim term “the biomolecule” that is affixed to a linker site and recited in the body of each of these claims. *See* Ex. 1001, 26:11–18 (claim 1), 28:13–22 (claim 48), 28:39–56 (claims 53 and 54).

Claims must also be read in view of the specification of which they are an integral part. *See Phillips*, 415 F.3d at 1315. The Specification of the ’441 patent, like the language of the preamble itself, repeatedly refers to the detection apparatus as identifying a “single” biomolecule. *See, e.g.*, Ex. 1001, 1:67–2:6 (stating devices should be capable of sequencing *single* molecules), 2:16–17 (stating each sequencing reaction uses a *single* molecule as the template), 3:55–4:16 (stating each optical detection apparatus 20 may be operated independently to detect and identify a *single* biomolecule, and by integrating a huge number of optical detection apparatuses 20 on substrate 10, a huge number of *single* biomolecules can be

detected and identified in parallel). Accordingly, both the claim language and the Specification support Patent Owner's position that the preamble is limiting.

We find Petitioner's argument that the preamble is not limiting and that one can "identify" a single molecule from a signal from multiple copies of the same molecule is inapposite when the challenged claims are read in light of the Specification of the '441 patent. *See* Reply 3–5. Although such a method for identification of a molecule exists and, as Petitioner identifies, is done by the Illumina sequencing platform, that is not what is described in the Specification of the '441 patent and the claims at issue here. *See Phillips*, 415 F.3d at 1317 ("However, while extrinsic evidence 'can shed useful light on the relevant art,' we have explained that it is 'less significant than the intrinsic record in determining "the legally operative meaning of claim language.'") (quoting *C.R. Bard, Inc. v. U.S. Surgical Corp.*, 388 F.3d 858, 862 (Fed. Cir. 2004)).

The Specification of the '441 patent describes a bioassay system in which a very large number of the apparatuses described and claimed in the '441 patent, in some embodiments over 10,000, are used in parallel "to determine, for example, the sequence of a genome or the profile of expressed genes in a tissue sample with high throughput." Ex. 1001, 3:58–65. In such a bioassay system, the Specification indicates that "[e]ach optical detection apparatus may sense the existence of *a* fluorophore on the *single* molecule by detecting photons emitted from the fluorophore." *Id.* at 3:60–61 (emphasis added). The Specification of the '441 patent contemplates running myriad optical detection apparatuses in parallel to detect a *single* or individual biomolecule in each such apparatus. The fact

that other “Sequencing Modalities” are mentioned that may be used with the claimed apparatus of the ’441 patent, and which involve the detection of signal from multiple molecules of the same species, does not undercut this fulsome description of what is disclosed in the ’441 patent. *See* Reply 8–11.

Petitioner also points to dependent claims 26 and 30 as supporting its view that the challenged claims encompass detection of signal from multiple biomolecules of a particular species at a linker site. Claim 26 depends from claim 16, which expressly claims a method of sequencing *a* nucleic acid, comprising the steps of affixing *one* nucleic acid molecule to the linker site of the apparatus of claim 1 and performing nucleic acid sequencing of that nucleic acid molecule. Ex. 1001, 26:59–64. Claim 26 adds a step to the method of claim 16 “wherein the nucleic acid is amplified at the linker site before nucleic acid sequencing.” *Id.* at 27:20–21. The term “amplified” is used nowhere else in the ’441 patent.

Dr. Neikirk opines that claim 26 “expressly contemplates detecting a signal from *multiple* molecules of the same species.” Reply 5 (citing Ex. 1003 ¶ 60). Dr. Neikirk opines:

Those of skill in the art understand that this [amplification of the nucleic acid at the linker site before sequencing] refers to a process in which a single molecule is first located at the linker site and then multiple copies of that molecule are generated. Thus, in this case, the signal is detected from multiple molecules of the same species.

Id.

We do not find Dr. Neikirk’s explanation here to be persuasive. Dr. Neikirk, who is admittedly not well-versed in molecular sequencing,⁵ does not explain how “amplification at the linker site” in the context of

⁵ *See* Ex. 2009, 89:8–10; 30:22–25; 37:21–22; 38:14–15.

claim 26 in which only one nucleic acid molecule has been affixed to the linker site would mean that a signal is detected from multiple molecules of the same species. In particular, even if multiple copies of the molecule are made through amplification, claim 26 (by dependency on claim 16) recites that only “one nucleic acid molecule” is affixed to the linker site. There is no further indication in the claim language concerning what is done with the amplified nucleic acid molecules.

Dr. Neikirk’s opinion assumes that the molecules made through amplification must be affixed to the same linker site as the original nucleic acid molecule and detected, an assumption that is not supported by the record evidence. Moreover, Patent Owner explains how the claimed amplification means that the apparatus is still only detecting signal from the single affixed molecule. Resp. 10 (“But as Dr. Harris explains, such amplification does not imply that multiple copies of a nucleic acid molecule are then affixed to one linker site for detection as an ensemble—particularly given that claim 26 depends from claim 16, which expressly requires performing nucleic acid sequencing of ‘one nucleic acid molecule’ affixed to the linker of the apparatus of claim 1.”) (citing Ex. 2008 ¶¶ 47–48). We credit Patent Owner’s interpretation over Petitioner’s because it more closely aligns with the surrounding claim language and Specification, as explained above.

Petitioner states that claim 30 that recites “affixing *one or more* biomolecule to the linker site of the apparatus of claim 1,” expressly contemplates multiple molecules at the linker site. Reply 7. Petitioner does not cite to any testimony from Dr. Neikirk to support this argument.

We agree that claim 30 contemplates using the apparatus of claim 1 in a method where more than one molecule is affixed, but that does not mean that the apparatus of claim 1 is incapable of “identifying a single biomolecule,” as its preamble recites. Indeed, the fact that Patent Owner uses the phrase “a single biomolecule” in claim 1, but “one or more biomolecule” in claim 30, suggests their meaning is not coextensive, further evidencing that claim 1 should not be construed to encompass an apparatus that can only identify multiple biomolecules.

Accordingly, we determine that the preambles of claims 1, 48, and 53–55 are limiting, require an apparatus capable of identifying a single biomolecule, and provide antecedent basis for “the biomolecule” as used in the body of the independent claims, which should be read as “the single biomolecule” introduced in the preamble.

In our Institution Decision, we also addressed the scope of the claims as they relate to the linker site limitations, which we have touched on in our discussion of claim 30 above. These limitations require generally “a linker site formed over the light detector, the linker site being treated to affix the biomolecule to the linker site.” *See* Ex. 1001, 26:14–15 (claim 1), 28:16–17 (claim 48), 28:42–43 (claim 53), 28:51–52 (claim 54) (replacing “to the linker site” with “to it”), 28:63–64 (claim 55). In our Institution Decision, we stated:

The challenged apparatus claims recite a structure that includes a linker site being treated⁶ “to affix the biomolecule” thereto, the apparatus thereby being capable of affixing “the

⁶ We noted in our Institution Decision that the Specification does not define the term “treated,” but does state that “[a]ppropriate means for affixing nucleic acids to a solid support are well known in the art.” Ex. 1001, 11:8–9.

[single]⁷ biomolecule” to the linker site. But how the claimed apparatus is intended to be employed does not differentiate the claimed apparatus from a prior art apparatus if the prior art teaches all the structural elements of the claim. *Hewlett-Packard Co. v. Bausch & Lomb Inc.*, 909 F.2d 1464, 1468 (Fed. Cir. 1990) (“[A]pparatus claims cover what a device *is*, not what a device *does*.”). As the Federal Circuit has recently reiterated, this means that a prior art reference “may anticipate or render obvious an apparatus claim—depending on the claim language—if the reference discloses an apparatus that is reasonably capable of operating so as to meet the claim limitations, even if it does not meet the claim limitations in all modes of operation.” *ParkerVision, Inc. v. Qualcomm Inc.*, 903 F.3d 1354, 1361 (Fed. Cir. 2018); *see also* Ex. 1001, 1:67–2:1 (“the devices should be ‘capable of’ sequencing single molecules”).

Although the challenged apparatus claims are “capable of” affixing a single biomolecule to the linker site, the structure of the apparatus is not limited to being capable of affixing *only* a single biomolecule to the linker site. This is evident, at least with respect to claim 1, by the recitation of claim 30, *viz.*, “affixing one or more biomolecule[s] to the linker site of the apparatus of claim 1.” Ex. 1001, 27:32–33; *see also Phillips*, 415 F.3d at 1314 (“the claims themselves provide substantial guidance as to the meaning of particular claim terms”). We further note that all of the challenged apparatus claims include the open ended transitional phrase “comprising,” and thus do not limit the claimed structure to a linker site that is capable of affixing *only* a single biomolecule for identification thereof. *See CIAS, Inc. v. Alliance Gaming Corp.*, 504 F.3d 1356, 1360 (Fed. Cir. 2007) (“In the patent claim context, the term ‘comprising’ is well understood to mean ‘including but not limited to.’”).

Dec. 10–11.

⁷ For the purpose of the Institution Decision, we interpreted the claim phrase “to affix *the* biomolecule” as referring to “a single biomolecule.”

Patent Owner challenges our preliminary determinations concerning how to construe the linker site limitations. Patent Owner asserts that a person of ordinary skill in the art “would have understood that the ‘linker site being treated to affix the [single] biomolecule’ limitation describes a structural configuration, not merely a capability.” Resp. 12 (citing Ex. 2008 ¶ 54). Patent Owner argues further that the transitional phrase “comprising,” does not nullify the express recitation of identifying a single biomolecule. *Id.* at 13–14 (citing *On Demand*, 442 F.3d at 1343–1344).

Petitioner agrees with our initial assessment set forth above. *See* Reply 20–22. Petitioner states: “The claim language here recites ‘apparatus **for identifying** a single biomolecule’ and ‘linker site being treated to affix the biomolecule to the linker site.’ Such language mirrors what the Federal Circuit has found to recite capability as opposed to configuration.” *Id.* (citing *ParkerVision*, 903 F.3d at 1362).

We agree with Petitioner that our original construction is correct. In examining the claim language to determine whether it recites a mere capability to carry out a function versus a particular configuration to carry out the function. *See ParkerVision*, 903 F.3d at 1361. As the Federal Circuit stated in *ParkerVision*, “[t]he language used in the claims is critical to deciding on which side of the line the claims fall.” *Id.*

Here, the claims describe the required function of “identifying a single biomolecule” using an apparatus with (1) a substrate having a light detector and (2) a linker site formed over the substrate, being treated to affix the biomolecule to the linker site, with the linker site a distance of less than or equal to 100 micrometers from the light detector. *See* Ex. 1001, 26:11–18 (claim 1). The structure described includes a substrate with a light detector

and a linker site over the substrate and no more than 100 micrometers away from the light detector. No specific structure is delineated in the claims for the linker site that would restrict use of the site for only a single biomolecule. The only requirement in the claims is the linker site *being treated* to affix the single biomolecule to it. Such a limitation is more akin to the types of claims that the Federal Circuit has deemed to be claim language reciting a capability versus a particular configuration. *See Ball Aerosol & Specialty Container, Inc. v. Limited Brands, Inc.*, 555 F.3d 984, 987 (Fed. Cir. 2009) (finding noninfringement where claims recited a configuration in which protrusions needed to be “resting upon” a cover and alleged infringing device was only reasonably capable of such a configuration); *Ericsson, Inc. v. D-Link Sys., Inc.*, 773 F.3d 1201, 1217 (Fed. Cir. 2014) (finding claims at issue recite capability such as “a processor *for arranging information* for transmission . . . which identifies a type of payload information”).

Therefore, we conclude that although the challenged claims requiring an apparatus for “identifying a single biomolecule” are “capable of” affixing a single biomolecule to the linker site, we remain persuaded that the structure of the apparatus is not limited to being capable of affixing *only* a single biomolecule to the linker site.

2. *Formed Over*

Patent Owner argues that a person of ordinary skill in the art would have understood “‘a linker site formed over the light detector’ to at least require that the linker site be configured directly or indirectly on top of, upon, or above and, as some dictionaries articulate it, ‘at a higher level or layer than,’ . . . the light detector.” Resp. 16–17 (citing Ex. 2008 ¶¶ 61–62;

Ex. 2012; Ex. 2013). Patent Owner argues that Figure 2 of the '441 patent depicts linker site 220 in a layer of the device above light detector 210. *Id.* at 17–18 (citing Ex. 1001, Fig. 2, 5:33–35; Ex. 2008 ¶¶ 62–63). Patent Owner further argues that Figures 1–4 support this interpretation, where a person of ordinary skill in the art would have understood formed over as meaning “in a layer above.” *Id.* at 18 (citing Ex. 2008 ¶¶ 64–65).

Petitioner contends that “formed over” should not be construed to require either a “layer” or positioning “above.” *See* Reply 12–14. For example, Petitioner asserts that the Specification does not require that the light source be in a layer, and instead discloses embodiments in which “the excitation light source may be *stand alone* from optical detection apparatuses 20 or bioassay substrate 10” or “the excitation light source may be integrated with optical detection apparatuses 20 or bioassay substrate 10.” *Id.* at 13 (citing Ex. 1001, 7:4–9). Petitioner contrasts claim 49 which expressly recites a light emitting layer. *See id.* (citing claim 49). Petitioner further contends that the claims do not recite the term “above,” and that interpreting the “formed over” to require “above” “would exclude from the scope of the claims disclosed embodiments simply if they are turned upside down.” *Id.* at 13–14 (citing Ex. 1058, 142:15–22, 148:14–15, 154:8–14, 154:23–155:6, 155:10–15).

In reviewing the totality of the claim language describing the apparatus for identifying a single biomolecule in light of the Specification of the '441 patent, it appears that the term “formed over” describes the close proximity of the linker site and the light detector so that the light detector can collect light emitted from the biomolecule and the relative positions of the linker site as over or above the light detector. *See, e.g.*, Ex. 1001, Abst.,

2:30–41. For instance, in the description of Figure 2 set forth above, *see supra* Section I.D., optical detection apparatus 20 is described as including a “light detection 210 formed on substrate 10, and a linker site 220 formed over light detector 210” and shows the linker site placed above the light detector. Ex. 1001, 4:55–57.

The Specification of the '441 patent makes clear that the term “over” means at a higher level vertically or above. For instance, in discussing Figure 3 set forth below, the Specification of the '441 patent states:

Referring to FIG. 3, there is illustrated a sectional view of optical detection apparatus 20 in accordance with one embodiment consistent with the present invention. As shown in FIG. 3, blind sheet 230 is *formed over* light detector 210 and *vertically spaced apart* from light detector 210 by distance H1. Blind sheet 230, which has a thickness T, includes pinhole 235 having a radius R1 (i.e., one-half of diameter D1). In this embodiment, linker site 220 may be *formed in* pinhole 235 to bind with a biomolecule (not shown).

Ex. 1001, 5:65–6:6 (emphases added).

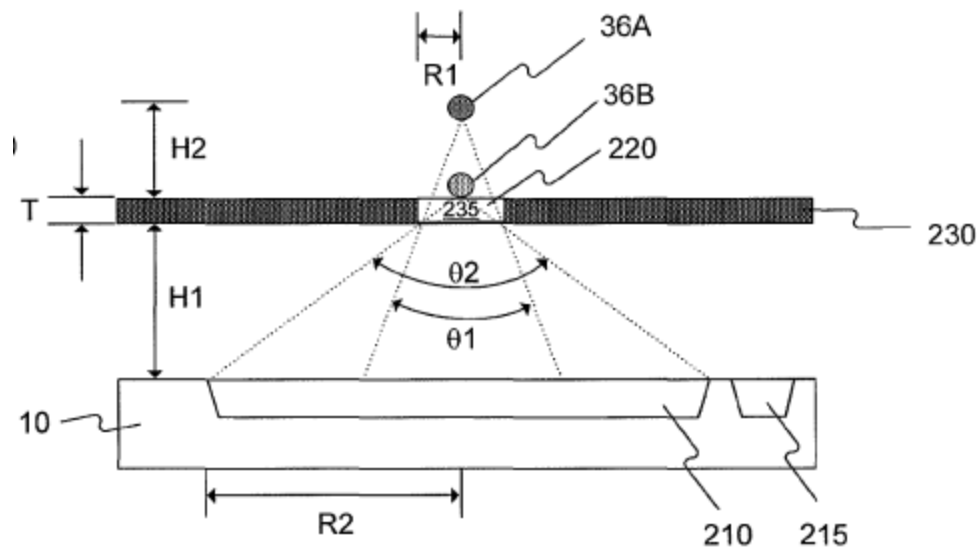


Fig. 3

The Specification of the '441 patent further highlights the meaning of “over” by juxtaposing it with forming “under.” For instance, in describing optical detection apparatus 20, the Specification of the '441 patent makes a distinction between a component “formed over” or “formed under.”

Optical detection apparatus 20 may further include a filter layer 240 (optional) and a microlens 250 (optional) between light detector 210 and blind sheet 230. Although FIG. 2 shows filter layer 240 is *formed over* microlens 250, it is appreciated that filter layer 240 may be *formed under* microlens 250.

Ex. 1001, 5:31–35 (emphases added). Finally, the Specification of the '441 patent further highlights that “over” means above in describing a method for manufacturing a bioassay system.

In this embodiment, a filter layer 240 is *formed over* the *upper surface* of light detectors 210 and control circuits 215. Global planarization process is applied to the *upper surface* of light detectors 210 and control circuits 215 before forming filter layer 240. Filter layer 240 includes a plurality of sublayers. In this embodiment, filter layer 240 is formed by first depositing a sublayer having a higher refractive index *over* the planarized *upper surface* of light detectors 210 and control circuits 215.

Id. at 19:47–55 (emphases added).

Therefore, we determine that “formed over” as used in the challenged claims means “placed above.”⁸

D. Anticipation by Choumane

Petitioner asserts that claims 1–5, 9, 48, and 52–56 are anticipated by Choumane. Pet. 8. Petitioner provides an analysis of how each claim

⁸ We do not see any description that restricts the linker site in a “layer.” Therefore, we do not include this in the construction of the term “formed over.”

limitation is taught by Choumane. *Id.* at 13–41. Petitioner relies on the declaration of Dr. Neikirk in support of its positions. *See generally* Ex. 1003.

Patent Owner asserts that Choumane teaches analyzing multiple copies of a molecule together, not identifying one molecule individually. Resp. 18–29. Patent Owner further asserts that Choumane does not disclose a linker site being treated to affix a single biomolecule, and Choumane does not disclose the linker site spaced apart from the light detector by a distance of less than or equal to 100 micrometers. *Id.* at 35–38. Patent Owner also presents specific arguments as to claims 9 and 48. *See id.* at 38–41. Specifically, Patent Owner argues that Choumane does not disclose a solid angle being greater than or equal to 0.8 SI steradian (claim 9), or an excitation light source formed over the substrate (claim 48). *See id.*

1. Choumane (Ex. 1006)

Choumane discloses that “[a] biosensor with integrated detection comprises a substrate for supporting chromophore elements and a set of photodetectors intended to capture the light emitted by the chromophore elements in response to light excitation, the set of photodetectors being associated with the substrate and forming a unitary assembly with it.” Ex. 1006, 1:4–7.

Figure 1 of Choumane depicts such a biosensor, and is reproduced below:

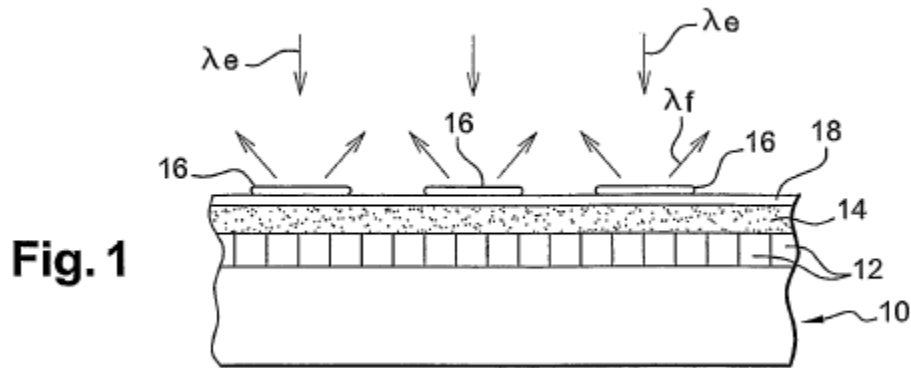


Figure 1 above illustrates a partial schematic sectional view of a biosensor. Ex. 1006, 7:16–17.

As shown above, Figure 1 illustrates a biosensor including matrix set 10 of photodetectors 12, on which is deposited absorbent layer 14 intended to absorb light radiation of chromophore elements located on areas or spots 16 of the surface of the biosensor. Ex. 1006, 8:9–12. Choumane discloses that “[a]bout 80% of the light flux emitted by the chromophore elements pass through the absorbent layer 14 and are picked up by the photodetectors 12.” *Id.* at 8:17–19. Choumane also discloses that, “[i]n the general case, a functionalization layer 18 is formed on the upper surface of the layer 14 on which the biological probes are deposited . . . [to allow] the fixation of the probes.” *Id.* at 11:3–5.

Choumane further discloses that “absorbent layer 14 may be formed of a single layer of absorbent material, or of multiple superimposed absorbent layers of different natures.” Ex. 1006, 8:21–22. Choumane discloses that “small” spacing between chromophores and photodetectors reduces parasitic crosstalk signals. *Id.* at 9:3–11. Choumane also discloses:

Due to the absence of an imaging element between the chromophores and the photodetectors, the same photodetector can receive light signals from different points or zones, which generates a parasitic crosstalk signal, the greater the spacing or

vertical distance between these points or zones, and the photodetector is large. In a preferred embodiment of the invention, this spacing is small and the spurious signal is reduced to a minimum. For example, the diameter of these points or zones is 400 μm and their spacing with the photodetectors is 10 μm , so that the spurious signal is minimal. When their spacing with the photodetectors is greater and reaches 100 μm , the crosstalk signal can be large and likely to reduce the sensitivity of the detection. In this case, computer processing for deconvolution of the parasitic image, well known to those skilled in the art, makes it possible to recover the useful signal by eliminating the parasitic signal.

Id. at 9:3–13.

Choumane discloses that biological probes are deposited on the surface of absorbent layer 14 to form spots 16. Ex. 1006, 10:16–17. The spotting composition may include a surfactant buffer liquid due to the hydrophobic nature of absorbent layer 14. *Id.* at 10:17–21. Choumane also discloses that “spots 16 have dimensions of 100 to 400 μm , . . . the absorbent layer 14 has a thickness of the order of 10 μm , the photodetectors 12 have unit dimensions of the order of 10 μm , the spots 16 therefore covering one or more tens of photodetectors.” *Id.* at 10:23–11:2.

Figure 8 of Choumane depicts a biosensor, and is reproduced below:

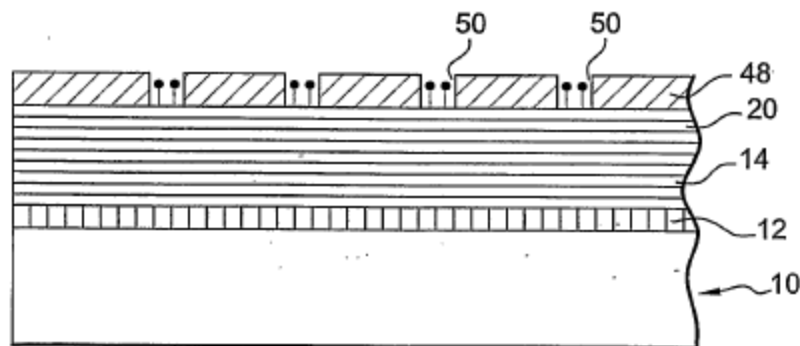


Fig. 8

Figure 8 above illustrates a schematic sectional view of a biosensor. Ex. 1006, 8:7–8. Specifically, Figure 8 illustrates a biosensor with matrix set 10 of photodetectors 12 with absorbent layers 14 deposited thereon. Ex. 1006, 14:13–19. The biosensor further includes layer 20 comprising a Bragg mirror (formed of thin layers of materials that are transparent at the chromophore’s emission wavelength) or an interference filter (formed for example of superimposed thin layers of polymers) deposited on absorbent layers 14. *Id.*; *see also id.* at 3:20–22.

As shown in Figure 8 above, Choumane discloses providing an opaque layer, such as a metal film 48, on rejection filter 20.⁹ Ex. 1006, 16:20–21; *see also id.* at 4:7–9. Film 48 has “very small openings 50, of a dimension less than the wavelength of light emitted by chromophores. These openings delimit very small observation volumes (for example with a diameter of 150 nm) for the detection and observation of individual chromophores. These openings also amplify the light emitted by the chromophores in their immediate vicinity.” *Id.* at 16:21–17:2.

Choumane also discloses a step-by-step process for manufacturing a biosensor. Ex. 1006, 13:11–12, 13:16–22.

2. Analysis

a) Claim 1

Petitioner argues that Choumane discloses the limitations of independent claim 1, and relies on the Neikirk Declarations as support. Pet. 13–25 (citing Ex. 1003 ¶¶ 60–93); Reply 15–23 (citing Ex. 1059 ¶¶ 5–

⁹ Choumane discloses that a “rejection filter comprises, in combination with the absorbent filter, a Bragg mirror . . . or an interference filter” which covers “at least one absorbent layer which is deposited on the set of photodetectors.” Ex. 1006, 3:19–24.

9). Patent Owner argues that Choumane does not anticipate claim 1 because Choumane does not disclose an apparatus for identifying one molecule individually. Resp. 18–38.

(1) An apparatus for identifying a single biomolecule

Petitioner argues that Figure 8 of Choumane discloses a metal film 48 with very small openings 50 having diameters of roughly 150 nm for detecting and observing individual chromophores. Pet. 14 (citing Ex. 1006, 16:20–17:2). Dr. Neikirk attests that “[t]he individual chromophore is associated with a biomolecule (e.g. DNA) and detection of the chromophore leads to the identification of its associated biomolecule.” Ex. 1003 ¶ 61. Petitioner relies on Weisbuch to confirm this association between one chromophore and one biomolecule. See Pet. 14–18.

Patent Owner generally argues that Choumane does not disclose identifying one molecule individually, rather Choumane discloses detecting a signal from multiple molecules of the same species. Resp. 19. Patent Owner asserts that Choumane’s reference to detecting “individual chromophores” does not mention “any biomolecule, let alone detection of a chromophore leading to the detection of an associated biomolecule.” *Id.* at 21 (citing Ex. 2008 ¶¶ 75–76).

Patent Owner further argues that Choumane does not disclose an apparatus for identifying a “single biomolecule” because Choumane’s Figure 8, which is a cross-sectional view, discloses two biomolecules in a cross-section of opening 50 as shown in the annotated version of that figure reproduced below. Resp. 23–24 (citing Ex. 2008 ¶ 78).

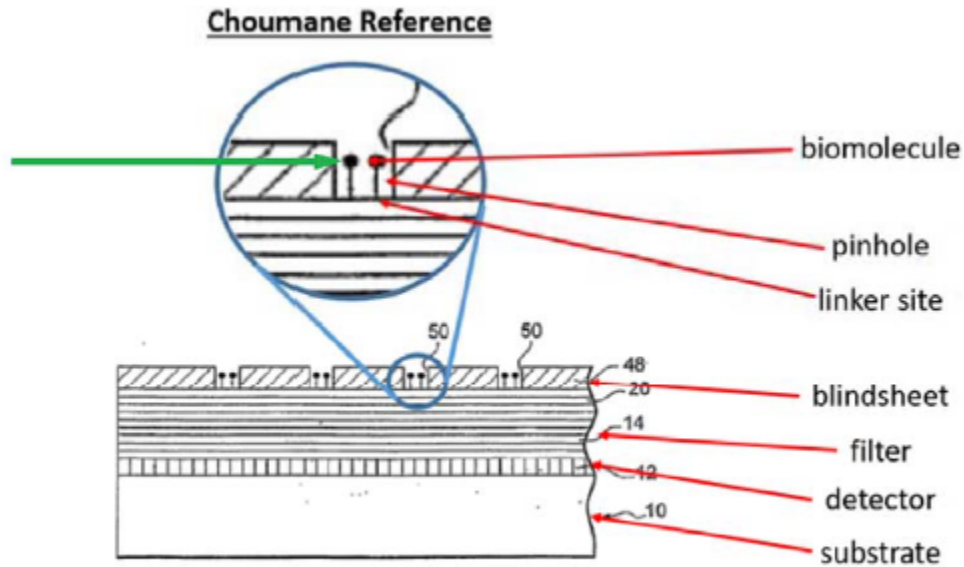


Fig. 8

This annotated version of Choumane Figure 8 schematically illustrates a biosensor with labeled parts, including a magnified window and arrows added by the parties. Resp. 24 (reproducing figure from Pet. 3). Patent Owner argues that Petitioner’s expert describes the stick-and-ball or pin-shape structures as biomolecules. *Id.* at 23 (citing Ex. 1003 ¶¶ 57–58; Ex. 2008 ¶ 78). Patent Owner argues that because opening 50 contains two biomolecules in a cross-sectional view, “a POSA would have understood that FIG. 8 at most discloses an apparatus that detects a signal from multiple biomolecules, not a single biomolecule.” *Id.* at 23–24.

Patent Owner further argues that Figure 8 is another example of pre-existing arrays including zones containing multiple probes specific for a particular molecular species. *See id.* at 25 (citing Ex. 1003 ¶¶ 66, 67; Pet. 17; Ex. 2008 ¶ 80). For example, Patent Owner contends that Petitioner’s reference to “spotting” indicates “a technique where multiple probes for target molecules are placed into ‘zones’ on the surface of an array, where ‘each zone contains probes specific for a particular molecular

species.” *Id.* at 25 (citing Ex. 1003 ¶ 67; Pet. 17; Ex. 1006, 6:5–13). Patent Owner asserts that Petitioner admits that Figure 8 is a variant of Figure 1, and Figure 1 discloses chromophore elements located in spots on the surface of the biosensor where “each spot has multiple probes for attaching multiple target molecules.” *Id.* at 26 (citing Ex. 1003 ¶¶ 56, 131, 52; Ex. 1006, 8:9–16; Ex. 2008 ¶ 81).

Petitioner responds that Choumane expressly discloses Figure 8 as an alternative embodiment for the detection and observation of individual chromophores. *See* Reply 15–16 (citing Ex. 1006, 16:20–24). Petitioner contends that a chromophore’s inherent purpose is to detect its associated biomolecule when employed in a biosensor. *Id.* at 16. Petitioner contends that Choumane describes the same chromophore as that used by the ’441 patent to detect biomolecules. *Id.* (citing Ex. 1006, 8:13–14; Ex. 1001, 21:10–11, 25:45–48).

Patent Owner also contends that “[a] POSA would have understood that Choumane does not even come close to disclosing an apparatus with single molecule or single chromophore sensitivity.” Resp. 27–29 (citing Ex. 2008 ¶¶ 84–85). Specifically, Patent Owner asserts that Choumane discloses a sensitivity of one chromophore/ μm^2 and photodectors with “unit dimensions of the order of 10 μm .” *Id.* at 28 (citing Ex. 1006, 3:6–9, 10:24–11:1). Patent Owner asserts that, a person of ordinary skill would have understood that Choumane’s photodetectors “have areas of at least about 78 μm^2 , assuming they are round” and “are therefore at best able to detect only as few as 78 chromophores each.” *Id.* (citing Ex. 2008 ¶ 86).

Petitioner responds that Choumane’s disclosure of “photodetector dimension on the order of 10 μm is merely an example of one size of

photodetector that may be implemented within the biosensor,” and Choumane discloses photodetectors having different sizes. *See* Reply 17–18 (citing Ex. 1006, 16:15–19). Petitioner argues that “a POSA would use a photodetector with dimensions that would enable the detection of an individual chromophore.” *Id.* at 18. Petitioner relies on Dr. Neikirk’s testimony in his rebuttal declaration that “the detector geometries disclosed in Choumane are, in fact, consistent with detecting signal from a single molecule.” *Id.* (citing Ex. 1059 ¶¶ 5–9).

We again note at the outset of our analysis that the claims require, at a minimum, an apparatus that is capable of identifying a single molecule, and thus, contemplates devices that in addition to being capable of identifying a single molecule may also identify multiple molecules. *See supra* Section II.C.1. Therefore, it is irrelevant to our analysis that Choumane may disclose detection of a signal from multiple molecules of the same species, so long as Choumane discloses an apparatus that is capable of identifying a single molecule. *See* Resp. 19–34.

It is undisputed that Choumane states that one alternative embodiment, depicted in Figure 8, has a biosensor with metal film 48 “having *very small openings* 50, of a dimension *less than the wavelength of light emitted by chromophores*. These openings delimit *very small observation volumes* (for example with a diameter of 150 nm) for the detection and observation of *individual* chromophores in solutions with high concentration.” Ex. 1006, 16:21–25 (emphases added).

In our view, Choumane unequivocally discloses an embodiment that is at least capable of identifying a single biomolecule by its express teaching that its biosensor can detect and observe *individual* chromophores. We

agree with and credit Dr. Neikirk’s testimony that an “individual chromophore is associated with a biomolecule (e.g. DNA) and detection of the chromophore leads to the identification of its associated biomolecule.” Ex. 1003 ¶ 61. Dr. Neikirk refers to Weisbuch that is cited in Choumane as also teaching that one chromophore is attached to a single biomolecule for detection. *See id.* ¶¶ 62–63. The ’441 patent also provides support for this truism stating that its biosensor “may sense the existence of *a* fluorophore on *the single* molecule by detecting photons emitted from *the* fluorophore.” Ex. 1001, 3:59–61 (emphasis added).

Patent Owner’s argument that Choumane’s written disclosure concerning Figure 8 does not mention an associated biomolecule or the identification of that biomolecule is not persuasive. The depiction in Figure 8 of two ball and stick figures that represent biomolecules with attached chromophores does not negate the specific, express teaching that Choumane’s apparatus is capable of detecting a single biomolecule. *See* Ex. 2008 ¶¶ 76–79 (relying on Figure 8 and discounting the written disclosure concerning Figure 8). A description of a figure can be relied on, in combination with the drawings, for what they would reasonably teach one of ordinary skill in the art. *In re Wright*, 569 F.2d 1124, 1127–28 (CCPA 1977).

Patent Owner’s argument that Choumane’s photodetectors in Figure 8 are sized to detect only as few as 78 chromophores does not address the full scope of the embodiments disclosed therein. Resp. 27–29. Ex. 2008 ¶ 85; *see* Ex. 1006, 10:16–11:2. As Petitioner points out, Choumane expressly discusses an embodiment depicted in Figure 7 that has “a matrix 10 of photodetectors 12a, 12b having different sizes” to accommodate “very weak

or very strong signals,” indicating that Choumane contemplates different sized photodetectors depending on the signal to be measured. Reply 17–18 (citing Ex. 1006, 16:15–16). Petitioner concludes, “[i]n an application of the biosensor, as described in FIG. 8, ‘for the detection and observation of *individual* chromophores in solutions with high concentration,’ a POSA would use a photodetector with dimensions that would enable the detection of an individual chromophore.” *Id.* at 18. Therefore, we are not persuaded by Patent Owner’s argument concerning the size of the photodetectors disclosed in Choumane.

We credit Dr. Neikirk’s testimony that the sensors of Choumane are capable of single molecule detection. Reply 18. Dr. Neikirk explains how if the photodetectors are on the rear face of the substrate or on the face opposite to that carrying the chromophores as disclosed in Choumane and polysilicon transfer grids on the photodetectors are avoided as Choumane suggests, the sensitivity of Choumane’s biosensor is capable of single molecule detection. Ex. 1059 ¶¶ 2–9. We find this testimony persuasive as it is consistent with Choumane’s disclosure, and credit it over the competing testimony from Patent Owner’s expert. *See* Ex. 2008 ¶ 86.

(2) a substrate having a light detector

Petitioner argues that Choumane discloses “a substrate having a light detector.” Pet. 18–20 (citing Ex. 1003 ¶¶ 69–71). Specifically, Petitioner argues that Figure 1 of Choumane “illustrates a biosensor comprising chromophore elements in spots 16 above an assembly 10 of photodetectors

12.”¹⁰ *Id.* at 19 (citing Ex. 1006, 8:9–16); *see also* Ex. 1006, 1:6–7 (“the set of photodetectors being associated with the substrate and forming a unitary assembly with it”). Petitioner also argues that Figure 8 of Choumane “illustrates an embodiment with an assembly 10 of photodetectors 12.” Pet. 19–20 (citing Ex. 1006, Fig. 8; Ex. 1003 ¶¶ 70, 71). Petitioner also cites to Choumane’s disclosure that the biosensor “comprises a substrate for supporting chromophore elements and a set of photodetectors,” and “the biosensor comprising a substrate intended to carry . . . a set of photodetectors associated with the substrate.” *Id.* at 18–19 (citing Ex. 1006, 1:4–5, 2:14–23).

We find that Petitioner has provided a sufficient showing as set forth above that Choumane teaches a substrate having a light detector. Patent Owner does not challenge Petitioner’s argument or evidence that Choumane teaches “a substrate having a light detector.” *See* Resp. 18–38.

(3) a linker site formed over the light detector

Petitioner argues that Choumane discloses “a linker site formed over the light detector.” Pet. 20–22 (citing Ex. 1003 ¶¶ 72–76). Specifically, Petitioner argues that Choumane “discloses predetermined zones on the biosensor where biomolecules for detection are attached” that correspond to the claimed linker site. *Id.* at 20 (citing Ex. 1006, 6:5–13; Ex. 1003 ¶¶ 73, 74). Petitioner refers to Choumane’s teaching that “biological probes can be

¹⁰ Petitioner asserts that the ’441 patent characterizes a “light detector” as a single photodetector or a group of photodetectors. Pet. 31 n.3 (citing Ex. 1001, 5:1–5). More precisely, we note that the ’441 patent states, “[i]n some embodiments, light detectors 210 may comprise a single photoconductive photon detector or a group of photoconductive photon detectors.” Ex. 1001, 5:1–3.

deposited directly on a rejection filter with thin layers which covers a set of photodetectors.” *Id.* (citing Ex. 1006, 3:3–9).

Referring to Figure 8 of Choumane, Petitioner argues that “biological probes are deposited onto the interference filter 20 in a pinhole opening 50 within the metallic film 48, which places the linker site over the light detector 12.” Pet. 21 (citing Ex. 1006, 6:5–13, 14:13–19, Fig. 8). Petitioner further argues that Figure 1 of Choumane depicts biological probes 16 that link target biomolecules to the surface of the biosensor, with spotting taking place over the assembly 10 of photodetectors 12. *Id.* at 22 (citing Ex. 1006, 10:16–11:5, Fig. 1).

We find that Petitioner has provided a sufficient showing, as set forth above, that Choumane teaches a substrate having a light detector. Patent Owner does not challenge Petitioner’s argument or evidence that Choumane teaches “a linker site formed over the light detector.” *See* Resp. 18–38.

(4) the linker site being treated to affix the biomolecule to the linker site

Petitioner argues that Choumane discloses “the linker site being treated to affix the biomolecule to the linker site.” Pet. 22 (citing Ex. 1003 ¶¶ 77–81). Petitioner cites to Choumane’s disclosure that when “this biosensor is produced, probes optionally comprising fluorescent markers are deposited in the liquid phase on specific zones . . . on the biosensor rejection filter.” *Id.* (citing Ex. 1006, 6:5–13, 14:13–19). Petitioner also cites to Choumane’s disclosure that “[o]ptionally, an aqueous buffer liquid containing a surfactant is used for depositing the probes on the rejection filter, the surface of which can be very hydrophobic.” *Id.* (citing Ex. 1006, 6:14–15). Petitioner argues that “[t]he location where a probe is deposited is a linker site because a biomolecule of interest will attach to that probe,

thereby ‘linking’ it to the apparatus.” *Id.* at 22–23 (citing Ex. 1003 ¶¶ 78, 79).

Patent Owner contends that Petitioner admits that Choumane discloses affixing plural “biological probes” in each pinhole opening in Figure 8 to capture “multiple molecules.” Resp. 35–36 (citing Pet. 21; Ex. 1006, Figs. 1, 8). Relying on its interpretation of Figure 8 set forth above, *see supra* Section II.D.2.a)(4)., Patent Owner argues “[t]his plainly does not result in a ‘linker site’ treated to affix a *single* biomolecule to be identified by the apparatus, as the claim requires.” *Id.* at 35.

We rely on our previous analysis that the claims require, at a minimum, an apparatus that is capable of identifying a single molecule. *See supra* Section II.C.1. We have determined that Choumane discloses such an apparatus. *See supra* Section II.D.2.a)(1); *see* Reply 21. We find that Petitioner has provided a sufficient showing as set forth above that Choumane teaches the linker site being treated to affix the biomolecule to the linker site.

(5) wherein the linker site is proximate to the light detector and is spaced apart from the light detector by a distance of less than or equal to 100 micrometers

Petitioner argues that Choumane discloses “the linker site is proximate to the light detector and is spaced apart from the light detector by a distance of less than or equal to 100 micrometers.” Pet. 23 (citing Ex. 1003 ¶¶ 82–93). Specifically, Petitioner argues that Choumane discloses minimizing crosstalk between photodetectors by limiting the distance between chromophores and their respective photodetectors to between 10 μm and 100 μm . *Id.* at 23–24 (citing Ex. 1006, 9:3–13 (“When [chromophore] spacing with the photodetectors is greater and reaches

100 μm , the crosstalk signal can be large and likely to reduce the sensitivity of the detection.”)).

Petitioner argues that Figure 1 of Choumane depicts a linker site separated from a detector by a 10 μm absorbent layer (and a thinner functional layer). Pet. 24 (citing Ex. 1006, Fig. 1, 10:16–11:2; Ex. 1003 ¶¶ 85–88). Petitioner also argues that Figure 8 of Choumane depicts a series of absorbent layers of roughly 10 μm thickness and Bragg mirror layers. *Id.* Petitioner also relies on Dr. Neikirk’s testimony to argue that “the POSA would understand that many of these layers correspond to Bragg mirrors, which are thinner than 10 μm such that the overall separation between the linker site and detector would, in fact, be substantially less than 100 μm .” *Id.* at 24–25 (citing Ex. 1003 ¶¶ 89–92). Petitioner further asserts that, according to Choumane, the biosensor may have “one or more thin absorbent layers” or “be formed of a single layer of absorbent material,” thereby disclosing that the biosensor of Figure 8 may have only a single absorbent layer, thereby decreasing the distance between the chromophore and the photodetector closer to the desired 10 μm distance. *Id.* at 25 (citing Ex. 1006, 5:9–12, 8:17–9:2, 9:3–13; Ex. 1003 ¶ 93).

Patent Owner asserts that Choumane seeks to limit crosstalk between zones of multiple molecules of one species, and “not from a single biomolecule, as required by claim 1.” Resp. 37 (citing Pet. 17–18; Ex. 2008 ¶¶ 106–109). Patent Owner further asserts that Petitioner calculates the distance between linker site and detector based on Choumane’s Figures; however, “Choumane expressly warns that “the relative dimensions of the various elements represented in FIG. 1 have not been respected in the drawing, for greater clarity.” *Id.* at 37 (citing Ex. 1006, 10:21–23).

Accordingly, Patent Owner argues that one of ordinary skill in the art “would not have concluded that the functionalization layer in FIG. 1 is thinner than the absorbent layer.” *Id.* at 37–38 (citing Ex. 2008 ¶ 113).

First, we note that Choumane expressly states, as Petitioner points out, that the spurious signal or crosstalk is minimal “when the spacing between the *points* or zones and the photodetector is 10 μm .” Ex. 1006, 9:6–9 (emphasis added). From this statement expressly referencing not only “zones,” but “points,” we find that Petitioner has shown sufficiently that Choumane teaches the limitation, “wherein the linker site is proximate to the light detector and is spaced apart from the light detector by a distance of less than or equal to 100 micrometers.” *See* Ex. 1003 ¶¶ 84–85 (stating “Choumane discloses a distance between the chromophore attached to the linker site and the light detector (i.e. photodetector) that encompasses a range between 10 μm and 100 μm .”).

Petitioner also refers to Figures 1 and 8 as illustrative of this express teaching. *See* Pet. 24–25. Dr. Neikirk explains that Figure 8 shows a distance between the photodetectors and chromophore that is under the preferred 100 μm limit. *See* Ex. 1003 ¶¶ 89–93. We find Dr. Neikirk’s explanation credible, and his testimony provides further evidence supporting what Choumane teaches as discussed above concerning the claim limitation—“wherein the linker site is proximate to the light detector and is spaced apart from the light detector by a distance of less than or equal to 100 micrometers.” *See In re Heck*, 699 F.2d 1331, 1332–33 (Fed. Cir. 1983) (stating a reference is relevant for all that it teaches).

(6) Conclusion

We determine that Petitioner has shown by a preponderance of the evidence that Choumane anticipates claim 1.

b) Claim 48

Independent claim 48 is an apparatus claim similar to claim 1, but which further includes the limitation “an excitation light source formed over the substrate.” Ex. 1001, 28:19. Petitioner argues that Choumane discloses the limitations of claim 48 for the same reasons as set forth with respect to claim 1. Pet. 33–34. Petitioner also argues that Choumane discloses the limitation of “an excitation light source formed over the substrate.” *Id.* at 33 (citing Ex. 1003 ¶¶ 127–131). Petitioner refers to Figure 1 to argue that “Choumane illustrates that the excitation light source λ_e is formed over the substrate comprising the matrix assembly 10 of photodetectors 12 such that the light impinges on the spots in a downward direction.” *Id.* (citing Ex. 1006, 8:9–16).

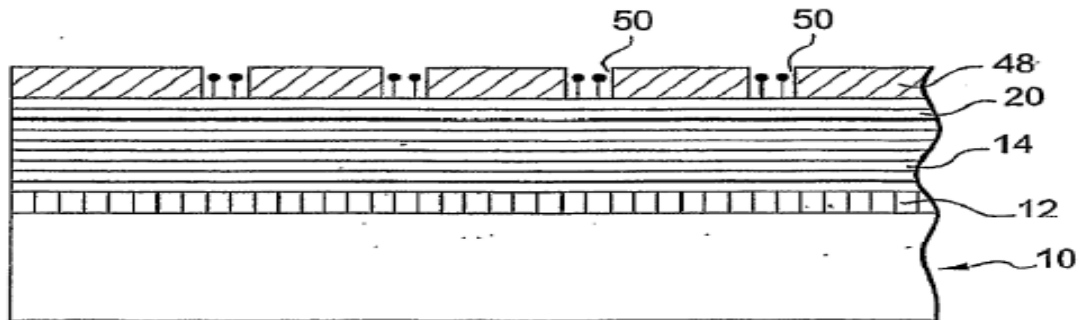
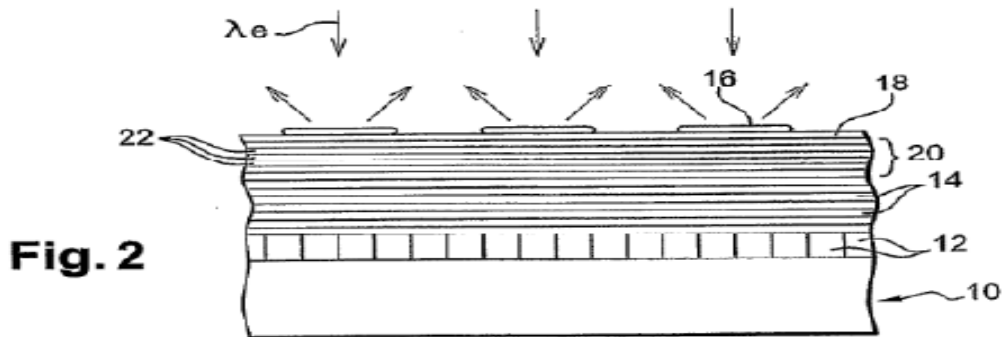
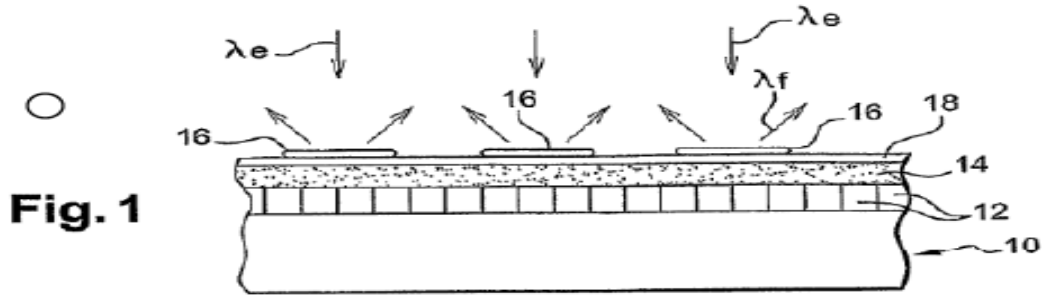
Petitioner also argues that Figure 8 of Choumane illustrates an embodiment having the same general structure as the embodiment shown in Figure 1, including a light source formed over the substrate that houses the light detectors. Pet. 34. As explained by Petitioner and supported by the testimony of Dr. Neikirk, Figure 8 illustrates a light filter (14) above the photodetectors 12 and substrate 10, a configuration that only makes sense if the excitation light source is formed over the substrate for the biomolecules. *Id.* (citing Ex. 1003 ¶¶ 128–131).

Patent Owner reiterates the arguments presented against claim 1 with respect to claim 48. *See* Resp. 40–41. Patent Owner further argues that Choumane does not disclose an excitation light source “formed over” the

substrate. *Id.* Specifically, Patent Owner argues that Choumane’s Figure 8 does not “depict any structure as a source for any excitation light,” nor does it include downward arrows indicating impinging light in Figure 1, relied on by Petitioner. *See id.* (citing Ex. 2008 ¶¶ 129–130).

We determine that Dr. Neikirk explains sufficiently how Choumane discloses an excitation light source formed over the substrate by comparing the configurations of Figure 1 and 2 that show an excitation light source above the substrate. *See* Ex. 1003 ¶¶ 127–131. Dr. Neikirk points out that Figures 1 and 2 show an assembly 10 of photodetectors 12 forming a unitary assembly with the substrate. *Id.* ¶ 128. Dr. Neikirk further testifies:

An excitation light source λ_e , as depicted in FIGS. 1 and 2, is formed over the substrate comprising an assembly 10 of photodetectors 12. The light source is necessarily formed over the substrate because the FIGS. show the light impinging in a downward direction on the linker sites:



Ex. 1003 ¶ 128.

As is shown in Figures 1 and 2 above, downward arrows directed at the spots 16 containing the chromophore elements show how light from above excites the chromophores. Ex. 1006, 8:9–19 (“About 80% of the light flux emitted by the chromophore elements pass through the absorbent layer 14 and are picked up by the photodetectors 12, the excitation light flux at the wavelength λ_e being absorbed by layer 14.”).

Patent Owner’s only argument that Choumane does not disclose “an excitation light source formed over the substrate” is the lack of downward arrows in Figure 8. We credit Dr. Neikirk’s testimony concerning how Figure 8, which shows a similarly constructed biosensor with the addition of metal film 48 with pinholes on the surface of the biosensor, would employ the same type of excitation light source as Figures 1 and 2. *See* Ex. 1003 ¶¶ 130–131. For instance, Dr. Neikirk testifies that the configuration of Figures 1 and 2 also applies to Figure 8 “because FIG. 8 is ultimately an addition of a metal film with pinholes onto the surface of the biosensors disclosed in Figs. 1 and 2.” *Id.* ¶ 131.

We determine that Petitioner has shown by a preponderance of the evidence that Choumane anticipates claim 48.

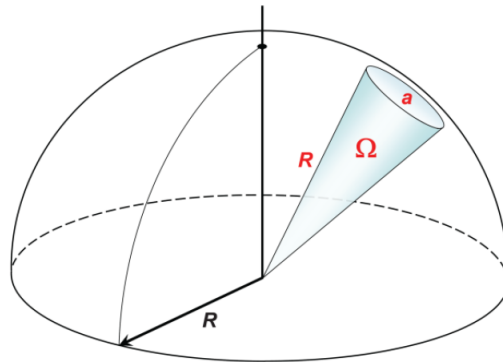
c) Claims 9, 53 and 54

Claim 9 depends from claim 1 and adds the limitation “wherein the light detector collects light from the biomolecule within a solid angle, the solid angle being greater than or equal to 0.8 SI steridian” (the “steridian limitation”).¹¹ Ex. 1001, 26:35–37. Independent apparatus claims 53 and 54 are similar to claim 1, but also include a similar limitation to claim 9— “wherein the light detector collects light emitted from the biomolecule within a solid angle of greater than or equal to 0.8 SI steridian” rather than the “less than or equal to 100 micrometers” limitation of claim 1. *See id.* at 28:39–56.

¹¹ The correct spelling of “steridian” is “steradian.” *See* Prelim. Resp. 43 n.3. However, we use the spelling “steridian” as used in the claims and by the parties.

Petitioner argues that Choumane discloses the other limitations of claims 53 and 54 for the same reasons set forth with respect to claim 1. Pet. 35–36. Petitioner also argues that Choumane discloses the steridian limitation. *Id.* at 36 (citing Pet. 31–33 (discussing dependent claim 9); Ex. 1003 ¶¶ 113–123).

Petitioner argues that “[s]olid angle Ω is a measure of the field of view (a) from a point” that “is expressed in a dimensionless unit called a steridian.” Pet. 31 (citing Ex. 1003 ¶¶ 114, 115). Petitioner argues that solid angle Ω is determined by the equation: $\Omega = a/R^2$ where “a” is the surface area of the spherical “cap” of the field of view, and “R” is the radius of the sphere. *Id.* at 31–32. Petitioner provides the following illustration of the units of that equation:



Id. The above illustration is a visual depiction of the solid angle Ω (steridian) based on the equation $\Omega = a/R^2$.

Petitioner argues that:

FIG. 1 of Choumane depicts a system in which the chromophore and detectors are separated from one another by roughly 10 μm . The chromophores are deposited as “spots” on the surface of the biosensor and, as Choumane explains, each spot “cover[s] one or more tens of photodetectors.” In other words, the minimum number of detectors covered by a spot is roughly 10.

Id. at 32 (emphasis omitted) (citing Ex. 1006, 9:3–13, 10:16–11:2).

Petitioner argues that “[c]onsidering a single molecule on the ‘spot’ and its nearest neighbor photodetector and next-nearest neighbor photodetectors yields a grid of nine photodetectors (each of which is roughly 10 μm in size) that will detect light from the single molecule.” *Id.*

Petitioner further argues that Choumane discloses

that the minimum number of photodetectors covered by each spot is about 10, and shows that the minimum width of the detection region for the single molecule is about 30 μm . A chromophore 10 μm away from the photodetectors that is emitting light onto a photodetector array of 30 μm diameter yields a solid angle of approximately 2.8 steradian, which is significantly greater than 0.8 steradian.

Id. at 32–33 (emphasis omitted) (citing Ex. 1003 ¶¶ 119–123).

Patent Owner argues that Choumane fails to disclose the steradian limitation because Choumane does not disclose an apparatus for identifying a single biomolecule, or any spot containing a single biomolecule.

Resp. 38–39. We are not persuaded by this argument because we have determined that Choumane discloses a device that is at least capable of identifying a single biomolecule. *See supra* Section II.D.2.a)(1).

Patent Owner also argues that the result of Petitioner’s calculation based on Figure 1 does not apply to Figure 8, the structure identified as the apparatus for identifying a single molecule, because Choumane does not disclose 10 μm spacing for Figure 8. Resp. 39 (citing Pet. 32–33; Ex. 2008 ¶¶ 118–122).

As Petitioner points out, Patent Owner does not dispute Dr. Neikirk’s calculations, just that Figure 8 does not teach the 10 μm spacing. *See Reply 23.* Petitioner points to the statement that in a preferred embodiment of Choumane, the spacing between the chromophore and the photodetector is 10

μm “so that the spurious signal is minimal.” *Id.* Petitioner also points to the statement that: “The absorbent layer 14 may be formed of a single layer of absorbent material, or of multiple superimposed absorbent layers of different natures to reduce the autofluorescence of this layer caused by the excitation light.” *See* Ex. 1006, 8:21–23. Petitioner concludes:

Given the foregoing, the distance between the chromophore and photodetector is effectively determined by the thickness of the absorbent layer; when there is only one absorbent layer with thickness of 10 μm, the distance is 10 μm. There is no reason, and Patent Owner does not provide any, to dispute that a minimum light distance of 10 μm, for example, is preferable for the embodiment of Choumane’s biosensors, including the embodiment characterized in FIG. 8.

Reply 25.

We agree with Petitioner. Choumane discloses closely related embodiments of biosensors with similar components for which Choumane expresses clear preferences as to the construction of the disclosed biosensor, including a preference for a 10 μm distance between the chromophore and the photodetector. We agree with Petitioner that this preference with the disclosure that absorbent layer 14 may be only one layer is readily applicable to the embodiment disclosed in Figure 8. *See In re Arkley*, 455 F.2d 586, 588 (CCPA 1972) (finding anticipation is shown when a reference clearly and unequivocally discloses the claimed invention without any need for picking, choosing, and combining various disclosures not directly related to each other by the teachings of the cited reference).

We determine that Petitioner has shown by a preponderance of the evidence that claims 9, 43, and 54 are anticipated by Choumane.

d) Claim 55

Petitioner argues that Choumane discloses “a method for manufacturing an apparatus for identifying a single biomolecule,” as recited in claim 55, including the preamble. Pet. 37–40 (citing Ex. 1003 ¶¶ 60–68, 146–151).

Petitioner argues that Choumane discloses the step of “forming a light detector and a control circuit on a substrate.” Pet. 38–39. Specifically, Petitioner argues that the light detector is formed on the substrate by referring back to its discussion of this limitation in connection with claim 1 and by citing to Choumane’s disclosure of step b of its fabrication process. *Id.* at 38–39 (citing Pet. 18–20; Ex. 1006, 13:16–18). Petitioner also argues that Choumane discloses a control circuit formed on the substrate. *Id.* at 39–40 (citing Ex. 1006, 12:23–13:2; Fig. 3; Ex. 1003 ¶¶ 152–156).

Petitioner argues that Choumane discloses the step of “forming a blind sheet having a pinhole over the substrate.” Pet. 40. Petitioner argues that Choumane’s “metallic film is a blind sheet and is located over ‘the set of photodetectors being associated with the substrate and forming a unitary assembly with it.’” *Id.* (citing Ex. 1006, 1:4–7; Ex. 1003 ¶¶ 157, 158; Pet. 26–27). Petitioner argues that “Choumane discloses that its metallic film layer has pinhole openings,” specifically very small openings 50 with, for example, a diameter of 150 nm. *Id.* at 27 (citing Ex. 1006, 16:20–17:2; Ex. 1003 ¶¶ 101, 102). Petitioner also argues that the remaining limitations of claim 55 are also disclosed in Choumane, referring back to its previous arguments. *Id.* (citing Pet. 20–25, 27–29).

Patent Owner asserts arguments for why claim 55 is not anticipated that are similar to those it asserted for claim 1. The same reasons that we

found that those arguments were not persuasive for claim 1 apply here. Therefore, we find Patent Owner's arguments are unavailing here.

We determine that Petitioner has shown by a preponderance of the evidence that claim 55 is anticipated by Choumane.

3. Dependent claims 2–5, 52, and 56

Petitioner asserts that Choumane discloses the limitations of dependent claims 2–5, 9, 52, and 56, and provides arguments and evidence supporting those assertions. Pet. 25–35, 41 (citing Ex. 1003 ¶¶ 94–107, 109–123, 133–136; Ex. 1006, 1:4–7, 3:3–9, 6:5–13, 8:9–9:2, 9:3–13, 10:16–11:2, 14:13–19, 16:20–17:5, 4:7–9, Fig. 8; Ex. 1001, 5:17–22). We determine that Petitioner has provided sufficient evidence to show by a preponderance of the evidence that dependent claims 2–5, 52, and 56 are unpatentable as anticipated by Choumane.

Patent Owner does not challenge those claims beyond the fact that they are ultimately dependent from one of the independent claims addressed above. *See* Resp. 38, 41, 46. For the same reasoning that we set forth for the challenged independent claims and our determination of the sufficiency of the evidence presented by Petitioner as to the additional limitations of these dependent claims, we also determine that Petitioner has shown by a preponderance of the evidence that claims 2–5, 52, and 56 are anticipated by Choumane.

E. Obviousness Grounds

Petitioner asserts that claim 1–5, 9, 48, and 52–56 are not only anticipated by Choumane (addressed above), but also obvious over Choumane and Weisbuch. Pet. 41–45 (citing Ex. 1003 ¶¶ 165–175). Petitioner asserts that “to the extent the preamble is deemed limiting and to

the extent Patent Owner contends that Choumane does not disclose this element,” claims 1–5, 9, 48, and 52–56 are obvious in view of Choumane and Weisbuch. *Id.* at 41. That is, “the only difference between Ground 2 and Ground 1 is that limitation [1A] (‘An apparatus for identifying a single biomolecule’) is treated as obviousness for ‘a single biomolecule’ rather than anticipation.” *Id.* at 41 (citing Ex. 1003 ¶¶ 165–175).

Patent Owner does not raise any argument as to why the claims are not rendered obvious over Choumane and Weisbuch except those arguments specific to Choumane that Patent Owner asserted in the anticipation ground based on Choumane. *See* Resp. 46–47 (addressing Choumane’s failure to disclose “the linker site being treated to affix the biomolecule” and “the linker site proximate to the light detector and spaced apart from the light detector by a distance of less than or equal to 100 micrometers.”).

Having found that Petitioner has shown by a preponderance of the evidence that Choumane anticipates claims 1–5, 9, 48, and 52–56, we rely on our previous discussion of Choumane for these claims addressing Patent Owner’s arguments in the anticipation challenge for the obviousness challenge here over Choumane and Weisbuch. *See Realtime Data, LLC v. Iancu*, 912 F.3d 1368, 1373 (Fed. Cir. 2019) (“[I]t is well settled that ‘a disclosure that anticipates under § 102 also renders the claim invalid under § 103, for ‘anticipation is the epitome of obviousness.’”) (internal citations omitted). We also agree with Petitioner that Weisbuch teaches an apparatus that is capable of identifying a single biomolecule. *See* Pet. 14–16, 41–42; Ex. 1007, Figs. 5, 8, ¶¶ 2–3, 5, 16, 48, 110; Ex. 1003 ¶ 166; *see also infra* Section II.E.1. (discussion of the teachings of Weisbuch that disclose an apparatus that is capable of identifying a single biomolecule). We also find

that a POSA would have had reason to combine Weisbuch that teaches identification of a single biomolecule with the technique of Choumane for identifying a single biomolecule for the reasons provided by Petitioner. *See* Pet. 41–45. Therefore, we determine that Petitioner has shown by a preponderance of the evidence that claims 1–5, 9, 48, and 52–56 would have been obvious over Choumane and Weisbuch.

Petitioner also asserts that claims 6, 9, 49–51, and 53–55 are unpatentable under 35 U.S.C. § 103 as obvious over Choumane and Weisbuch. Pet. 8, 41–59. Petitioner also asserts that claims 43–48 and 57–58 are unpatentable under 35 U.S.C. § 103 as obvious over Choumane in view of the knowledge or skill of a person of ordinary skill in the art. *Id.* at 8, 59–77.

We address these contentions below.

1. Weisbuch (Ex. 1007)

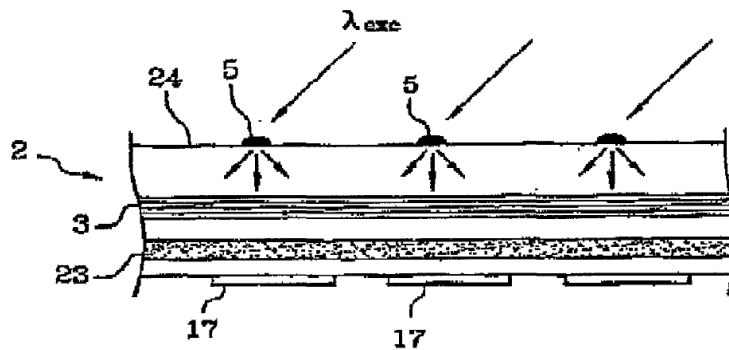
Weisbuch discloses a device for identifying the presence of a particular molecule (target) in a complex mixture by contacting the target with a probe fixed on a support or substrate. Ex. 1007 ¶ 5. The “target is detected when it couples to a fixed probe (or specific ligand) because the target carries a chromophore.” *Id.*; *see also id.* ¶ 48.

Weisbuch discloses a “device for supporting chromophoric elements . . . which can concentrate at least a portion of the light emitted by the chromophoric elements in a selected region of space, to allow it to be captured.” Ex. 1007 ¶ 15. The device may include a “substrate having an upper surface on which said chromophoric element is fixed and means for enhancing the quantity of light emitted by the chromophoric element

towards a collecting device.” *Id.* ¶ 16. The quantity of light may be enhanced by a microlens. *Id.* ¶ 17.

Weisbuch illustrates a biosensor in Figure 8, reproduced below.

Fig. 8



The illustration above depicts a cross-sectional view of a biosensor device. Ex. 1007 ¶ 57.

Figure 8 depicts substrate 2, carrying chromophores 5, multilayered reflective means 3, and absorbent layer 23 located between reflective layers 3 and photodetectors 17. Ex. 1007 ¶ 137. Weisbuch discloses that “reflective layers 3 ensure a maximum excitation and emission field at the chromophores and prevent the excitation light from propagating by simple transmission and by diffusion towards the photodetecting matrix 17.” *Id.* Absorbent layer 23 “has an absorption band and thickness selected to reduce crosscolor and/or enhance the signal-to-noise ratio at photodetectors 17.” *Id.*

Weisbuch also discloses a biosensor that includes a substrate and a multiplicity of holes “each intended to receive a target. The holes allow the chromophores to be placed in locations where the field of the resonator is a maximum.” Ex. 1007 ¶ 110. Weisbuch further explains that “[t]o collect the light emitted by chromophore 5 formed by interaction of a probe with the target lodged in a hole 11, a planar resonator 12 is formed around each hole

11.” *Id.* Weisbuch discloses receiving different probes in selected locations by fixing the probes to a support using addressing techniques known in the art. *Id.* ¶ 68. Weisbuch also discloses including microlenses located below each chromophore 5, which focus the emitted light toward the photodetector. *Id.* ¶¶ 73, 123, claim 1.

2. *Claims 6, 9, 49–51, 53, and 54*

Petitioner provides arguments and evidence supporting the obviousness of claims 6, 9, 49–51, 53, and 54 over Choumane and Weisbuch. Pet. 45–59.¹² Petitioner points to where Weisbuch teaches the additional limitations added by dependent claims 6, 9, and 49–51. *See id.* at 45–56; *see* Ex. 1003 ¶¶ 176–222. Petitioner further states:

There is a natural motivation to combine the microlens of Weisbuch with Choumane. Indeed, Choumane directly refers to Weisbuch, stating that an “object of the present invention is... to further improve the biosensor with integrated detection described in document WO 02/16912” (i.e. the WO filing of the Weisbuch reference). EX1006 at 2:12-13. Choumane further states that the structures in Weisbuch “have the advantage of improving the sensitivity of detection thanks to a very significant increase in the efficiency of the collection of the light emitted by the chromophores and in a reduction in the capture of the excitation light as well as a reduction in the parasitic fluorescence coming from the ambient environment.”

¹² Independent claims 53 and 54 were addressed with respect to the anticipation ground based on Choumane, and we found Choumane anticipated these claims. *See* Section II.D.2.c). Petitioner alleges that to the extent “Choumane fails to disclose identification of a single biomolecule . . . the combination of Choumane with Weisbuch still renders [claims 53 and 54] obvious for the additional reasons stated with respect to Ground 2.” Pet. 57, 58. Because we have found these claims anticipated by Choumane and did not address the contentions in the previous Choumane/Weisbuch obviousness ground for these claims, we will not further address this obviousness combination as to these claims here.

Id. at 1:17-2:4. Therefore, a POSA would be motivated to implement a range of advantageous elements from the Weisbuch system in Choumane. EX1003 ¶ 185.

Pet. 47.

Based on Petitioner argument and supporting evidence, we determine that Petitioner has shown by a preponderance of the evidence that claims 6, 9, and 49–51 would have been obvious over the combination of Choumane and Weisbuch. Patent Owner does not challenge Petitioner’s argument and evidence except to assert the same arguments with respect to how Choumane fails to teach the limitations of the independent claims from which claims 6, 9, and 49–51 depend. *See* Resp. 50–54. We have addressed these arguments in the anticipation ground. *See* Section II.D.2.

Therefore, we determine that Petitioner has shown by a preponderance of the evidence that claims 6, 9, and 49–51 are unpatenable over the combination of Choumane and Weisbuch.

3. *Claims 43–48*

a) *Claims 43–47*

Petitioner argues that claims 43–47 are directed to nothing more than the use of systems disclosed in claim 1 in various business configurations. Pet. 59. Petitioner specifically asserts the knowledge of a POSA that (1) a service requester would request, and receive, results from a service provider, (2) remunerative consideration can be provided for performing a service based on claim 1 of the ’441 patent, (3) the service requester and the service provider are mediated by a vendor, and (4) results can be produced in a country other than the U.S.A. *Id.* at 60–64 (citing Ex. 1003 ¶¶ 232–247).

Patent Owner does not dispute Petitioner’s evidence as to the knowledge of a POSA set forth above, and we credit Dr. Neikirk’s testimony

concerning the knowledge of a POSA as it relates to claims 43–47. *See* Ex. 1003 ¶¶ 232–247; Ex. 1006, 3:6–9 (discussing mass production of miniaturized and low cost biosensors). Patent Owner does again question whether Choumane discloses the apparatus of claim 1; arguments that we have previously addressed here. *See supra* Section II.D.2.a).

Therefore, we conclude that Petitioner has shown by a preponderance of the evidence that claims 43–47 would have been obvious over Choumane and the knowledge of a POSA.

b) Claim 48

As to claim 48, Petitioner relies on its arguments regarding anticipation of claim 1, and arguments that Choumane discloses the limitation of “an excitation light source formed over the substrate.” Pet. 64–67 (citing Pet. 13–25; Ex. 1003 ¶¶ 251–254; Ex. 1006, 8:9–16). Dr. Neikirk testifies:

To the extent one contends that the disclosure of light λ_e does not establish that the light source that emits λ_e is formed over the substrate, it would have been obvious to a POSA to put the light source for λ_e over the substrate. As a threshold matter, there needs to be a light source for λ_e . It would have been obvious to put that light source over the substrate – *i.e.*, obvious to try. In showing the light impinging on the linker sites in a downward direction, Choumane expressly instructs the skilled artisan to form the light source over the substrate.

Moreover, there are only a finite number of way of providing light λ_e incident to the device, the most obvious being from a light source formed over the substrate. I can think of no reason that a POSA would not have had a reasonable expectation of success of forming a light source over the substrate.

Ex. 1003 ¶¶ 253–254. Patent Owner raises arguments that we have previously addressed in the anticipation ground, and also responds specifically to the testimony set forth above. *See* Resp. 57–60. Specifically, Patent Owner asserts that Choumane discloses *le* as light, but not a source of the light or its structure. *Id.* at 59. Patent Owner also asserts that Dr. Neikirk’s testimony “merely addresses the position of the light source. Claim 48 requires not just that the excitation light source be ‘over’ the substrate, but that it be ‘*formed* over’ the substrate.” *Id.* at 59–60 (citing Ex. 2008 ¶¶ 199–200).

Although we have previously found that Choumane teaches the additional limitation of claim 48 and anticipates this claim, we also credit Dr. Neikirk’s testimony here concerning the knowledge of a POSA as it relates to claims 48. We disagree with Patent Owner that Dr. Neikirk’s testimony merely addressed the position of the light source and did not address whether the light source is “formed over” the substrate. As we have previously determined, “formed over” means “placed above.” *See supra* Section II.C.2. Using the drawings of Choumane to show the direction of the light that must come from a light source, we find that Dr. Neikirk sufficiently explained how a POSA would have found it obvious to place the light source over the substrate.

Therefore, we determine that Petitioner has shown by a preponderance of the evidence that claim 48 would have been obvious over Choumane and the knowledge of a POSA.

4. *Claims 57 and 58*

Petitioner asserts that dependent claims 57 and 58 (ultimately dependent from claim 55) are unpatentable as obvious over Choumane in view of the knowledge of a POSA.¹³ Pet. 67–77. As to claim 57, Petitioner argues that it is directed to the same apparatus recited in claim 2¹⁴ and, to the extent claim 57 includes any additional limitations, “those limitations are nothing more than basic photolithography steps that were well-known in the art long before the filing of the ’441 patent.” *Id.* at 67 (citing Ex. 1003 ¶ 256). As to claim 58 requiring “wherein the opaque layer comprises metal,” *see* Ex. 1001, 30:5–6, Petitioner cites to Choumane’s disclosures regarding a metallic opaque layer. Pet. 76–77 (citing Ex. 1006, 4:7–9, 16:20–17:2); *see* Ex. 1003 ¶ 265.

Patent Owner reiterates the same arguments concerning Choumane that we previously addressed in the anticipation ground, and does not dispute Petitioner’s evidence as to the knowledge of a POSA. *See* Resp. 60–61.

We credit Dr. Neikirk’s testimony concerning the basic photolithography steps that are well known to a POSA and would teach the limitations of 57 and 58, and we also find that Choumane itself discloses a metallic opaque layer as required by claim 58. *See* Ex. 1003 ¶¶ 264–291; Ex. 1006, 4:7–9, 16:20–17:2).

¹³ Petitioner cites to Ex. 1003 ¶¶ 256, 260, 264–270, 272–275, and 278–287 as well as several prior art references to establish the relevant knowledge of a POSA, particularly U.S. Patent No. 6,812,539 B1, issued Nov. 2, 2004 (Ex. 1023, “Rhodes”). *See* Pet. 67–77.

¹⁴ Claim 2 depends from claim 1 and recites “further comprising a blind sheet formed over the substrate, the blind sheet including a pinhole having a diameter, wherein the linker site is formed proximate to the pinhole.” Ex. 1001, 26:19–22.

Therefore, we determine that Petitioner has shown by a preponderance of the evidence that claims 57 and 58 would have been obvious over Choumane and the knowledge of a POSA.

III. CONCLUSION¹⁵

Based on the foregoing analysis of the arguments presented in the papers and the evidence submitted in support, we determine that Petitioner has shown by a preponderance of the evidence that claims 1–6, 9, and 43–58 are unpatentable.

IV. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that Petitioner has proven by a preponderance of the evidence that claims 1–6, 9, and 43–58 of U.S. Patent No. 7,767,441 B2 are unpatentable;

FURTHER ORDERED that, pursuant to 35 U.S.C. § 328(b), upon expiration of the time for appeal of this Decision, or the termination of any such appeal, a certificate shall issue canceling claims 1–6, 9, and 43–58 of U.S. Patent No. 7,767,441 B2; and

¹⁵ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner's attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. See 37 C.F.R. § 42.8(a)(3), (b)(2).

FURTHER ORDERED that, because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

In summary:

Claims	35 U.S.C. §	Reference(s)/Basis	Claims Shown Unpatentable	Claims Not shown Unpatentable
1–5, 9, 48, 52– 56	102	Choumane	1–5, 9, 48, 52–56	
1–5, 9, 48, 52– 56	103	Choumane, Weisbuch	1–5, 9, 48, 52–56	
6, 9, 49– 51, 53– 54	103	Choumane, Weisbuch	6, 9, 49–51	
43–48	103	Choumane in view of the skill of a POSA	43–48	
57, 58	103	Choumane in view of the knowledge of a POSA	57, 58	
Overall Outcome			1–6, 9, 43–58	

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